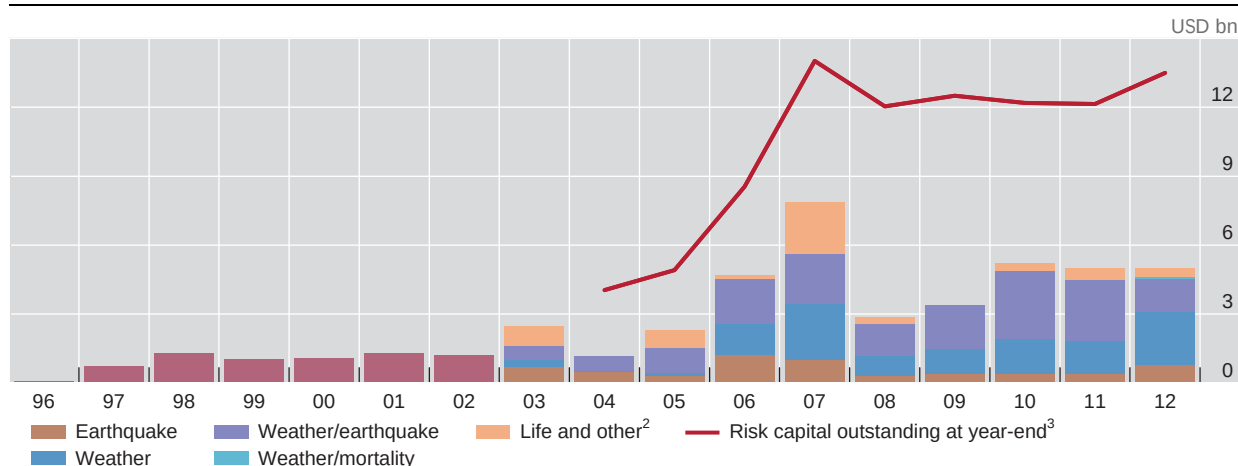


Catastrophe bond issuance by type of risk¹

Graph 8



¹ Data before 2003 are not broken down by type. ² Includes mortality, peril, life and worldwide risks. ³ Values are year-to-date, thus the value for 2012 is not final.

Sources: Artemis; Guy Carpenter.

to other bonds, the possibility of total loss is part of the arrangement from inception, and is compensated ex ante by a higher coupon.

Despite experts' high expectations, the catastrophe bond market has remained relatively small. Bond issuance has never exceeded \$7 billion per year, limiting the outstanding capital at risk to \$14 billion (Graph 8). Very few catastrophe bonds have been triggered to date. The 2005 Gulf Coast hurricanes activated payouts from only one of nine catastrophe bonds outstanding at the time (IAIS (2009)). Likewise, the 2011 Japan earthquake and tsunami triggered one known catastrophe bond, resulting in a payout of less than \$300 million. Payouts to reinsurers from these bonds are small when compared to the sum of insured losses (\$116 billion in 2005 and \$110 billion in 2011).

The global financial crisis has also dealt a blow to this market. The year 2008 saw a rapid decline in catastrophe bond issuance, reflecting generalised funding pressure and investor concern over the vulnerability of insurance entities. The crisis also demonstrated that securitisation structures introduce additional risk through linkages between financial entities. A case in point was the Lehman Brothers bankruptcy in September 2008. Four catastrophe bonds were impaired – not due to natural catastrophes, but because they included a total return swap with Lehman Brothers acting as a counterparty. Following Lehman's failure, these securitisation arrangements were no longer fully funded, and their market value plunged. Investors thus learned that catastrophe bonds are not immune to “unnatural” disasters such as major institutional failures.¹³

A further set of financial linkages arises with other financial institutions through cross-holdings of debt and equity. Insurance companies hold large positions in fixed income instruments, including bank bonds. At the same time, other financial entities own bonds and stocks in insurance companies. For instance, the two largest reinsurance companies stated in their latest (2011) annual reports that Warren

¹³ Following this episode, sponsors of catastrophe bonds employed other types of collateral arrangements in lieu of total return swaps. There has recently been a shift towards the use of government bonds as collateral.